

Recovering Uniform Factors

Exact Moment Fibres and Logarithmic Instability
Near the Fabius–Rvachev Law

Research article prepared for Vladimir Reshetnikov
AI-assisted mathematical draft; proofs and reproducible checks included

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Abstract

Let $X_a = \sum_{j \geq 1} a_j U_j$, where the U_j are independent uniform random variables on $[-1, 1]$ and a is a decreasing, nonnegative, summable sequence. The distribution of X_a determines a uniquely; this is an established identifiability theorem, not a new claim of this article. We investigate how much information survives finite moments and small errors in the distribution.

First, for every M , we construct a nonconstant real-analytic family through the dyadic Fabius–Rvachev spectrum that preserves its exact support and all moments through degree $2M + 1$. Second, Chebyshev level sets give explicit pairs a_n^+, a_n^- whose laws have equal moments through degree $2n - 1$ but satisfy

$$d_\infty(a_n^+, a_n^-) \geq \frac{1}{48n^2}, \quad \text{TV}(\text{Law}(X_{a_n^+}), \text{Law}(X_{a_n^-})) \leq 40n(6/7)^{2n-1}.$$

The laws converge in total variation to the Fabius–Rvachev law. A common normalization fixes their variance at $1/9$, keeps support inside $[-3, 3]$, and preserves uniform bounds on every derivative of their densities; the spectral lower bound then becomes $1/(96n^2)$. Consequently the inverse is not pairwise Hölder on any total-variation neighborhood of that law. Its local modulus is bounded below by a constant times $(\log(1/\varepsilon))^{-2}$, and estimation from N independent observations has a corresponding minimax lower bound of order $(\log N)^{-2}$. We give complete proofs, positive recovery results under finite-dimensional separation assumptions, and a reproducible exact-arithmetic verification suite. The exponent is a lower-bound exponent, not a proved optimal rate.

Status. The quantitative construction is presented as a proposed research contribution. Its priority has not been established by an exhaustive literature search. No claim to solve a named, previously published open conjecture is made. The proofs below are conventional and unrefereed; neither they nor the accompanying computations constitute a Lean formalization.

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1 The inverse question and its relation to ProveIt

1.1 From a forward construction to an inverse problem

The Fabius-function part of ProveIt develops the Rvachev up-function, infinite products of sinc functions, geometric-uniform sums, and their moment identities. Its project documentation explicitly identifies the two papers of Arias de Reyna as foundational sources [1, 2, 3]. The natural forward question is to determine a distribution, its transform, or its moments from the lengths of its independent uniform summands. Here we reverse that direction: *how reliably can those lengths be recovered from the distribution?*

This question has three different meanings. With the entire distribution known exactly, identifiability is already understood. With only finitely many exact moments, there can be infinitely many different answers. With an approximately known entire distribution, an inverse can be continuous and nevertheless fail every power-law error bound. The article separates these issues rather than using the word “recoverable” for all three.

Billey and Swanson prove identifiability for generalized uniform sums in a larger square-summable parameter space; see their Theorem 3.13 and Lemma 3.11 [4]. Their Theorem 1.15 gives a compact homeomorphic parametrization after allowing a Gaussian component in a suitable standardized space. We do not claim those results as new. The specific contribution here is an explicit, quantitatively ill-conditioned family inside the compact-support uniform-convolution class, localized near a distinguished smooth law already central to ProveIt.

Classical moment inversion is also known to be ill-conditioned. For example, Gerth, Hofmann, Hofmann, and Kindermann analyze conditional stability and failure of Hölder stability for the Hausdorff moment problem [5]. That general theory does not by itself produce the objects needed here: every parameter must be a genuine multiset of positive uniform half-lengths, the observable is the full probability law rather than just a sequence of noisy moments, and the laws must remain uniformly smooth. Our construction meets those constraints directly.

1.2 Notation and the main conclusion

Let $U_1, U_2, \dots$ be independent and uniform on $[-1, 1]$. For $L > 0$ set

$$\mathcal{A}_L = \left\{ a = (a_j)_{j \geq 1} : a_1 \geq a_2 \geq \dots \geq 0, \quad \sum_{j \geq 1} a_j \leq L \right\}.$$

Define

$$X_a = \sum_{j \geq 1} a_j U_j, \quad \mu_a = \text{Law}(X_a), \quad d_\infty(a, b) = \sup_{j \geq 1} |a_j - b_j|.$$

Zero coefficients are padding, not additional identifiable factors. The sum converges absolutely for every outcome. We use

$$\text{TV}(\mu, \nu) = \sup_{B \text{ Borel}} |\mu(B) - \nu(B)| = \frac{1}{2} \|f - g\|_1$$

when densities f, g exist. All logarithms without a subscript are natural. Write

$$a^0 = (2^{-1}, 2^{-2}, \dots), \quad \mu_0 = \mu_{a^0}, \quad f = \text{up}$$

for the density of μ_0 . In this normalization the support is $[-1, 1]$, $f(0) = 1$, and $\text{Var}(X_{a^0}) = 1/9$; these facts are proved in [section 3](#).

Definition 1.1. Let $\mathcal{K}$ be the subset of $\mathcal{A}_3$ consisting of spectra whose laws have variance $1/9$ and a density $g \in C_c^\infty(\mathbb{R})$ satisfying, for every integer $r \geq 0$,

$$\|g^{(r)}\|_\infty \leq 2^{r+1} \|f^{(r)}\|_\infty.$$

Thus $\mathcal{K}$ imposes a common support bound, an exact variance, and a fixed bound for each derivative order. It contains a^0 .

Theorem 1.2 (Quantitative instability near the dyadic law). *For every $n \geq 2$ there are explicit spectra $\tilde{a}_n^\pm \in \mathcal{K}$ with the following properties:*

$$d_\infty(\tilde{a}_n^+, \tilde{a}_n^-) \geq \frac{1}{96n^2}, \quad (1)$$

$$\text{TV}(\mu_{\tilde{a}_n^+}, \mu_{\tilde{a}_n^-}) \leq \varepsilon_n := \min\{1, 40n(6/7)^{2n-1}\}, \quad (2)$$

$$\text{TV}(\mu_{\tilde{a}_n^\pm}, \mu_0) \leq \sqrt{\frac{5}{12n}} + \frac{45}{16n}. \quad (3)$$

The two laws have identical moments through degree $2n - 1$ and different moments in degree $2n$.

For every $r > 0$ and every $C, \alpha > 0$, the estimate

$$d_\infty(a, b) \leq C \text{TV}(\mu_a, \mu_b)^\alpha$$

fails for some $a, b \in \mathcal{K}$ with both laws in the total-variation r -ball about μ_0 . Moreover, for all sufficiently small $\varepsilon > 0$, the local inverse modulus in that ball is at least

$$\frac{1}{11616[\log(100/\varepsilon)]^2}.$$

The proof occupies [sections 5 to 8](#). The important feature is not the numerical constant. It is the contrast between a polynomial separation of parameters and an exponentially small separation of full laws, under simultaneous smoothness and variance control.

1.3 What is established, proposed, and left unresolved

Statement	Status in this article
Cumulants and exact identifiability	Established background; re-derived with attribution to Billey–Swanson.
Finite moment fibres through a^0	A proved local construction; not advertised as a named open-conjecture solution.
Chebyshev pairs and explicit TV bound	Proposed quantitative contribution; proved for every n , independently of numerical tests.
Fixed-variance, uniformly smooth instability	Consequence of the explicit construction, with all normalization constants given.
Logarithmic statistical obstruction	A lower bound proved by a two-point argument, not an optimal minimax rate.
Optimal inverse modulus and pointwise stability at a^0	Not settled here; precise questions appear in section 11 .

The repository review was targeted, not an exhaustive audit of ProveIt. The source reference is pinned to the observed commit `74f7f5bdba1aa728b340509701a0fa4e45e8dbf3`; the accompanying source notes identify which documentation and outside literature were consulted.

2 Transforms, cumulants, and qualitative recovery

2.1 Support and the power-sum encoding

Proposition 2.1. *For $a \in \mathcal{A}_L$, put $S(a) = \sum_j a_j$ and $p_k(a) = \sum_j a_j^{2k}$. Then*

$$\text{supp } \mu_a = [-S(a), S(a)], \quad (4)$$

$$\phi_a(t) := \mathbb{E} e^{itX_a} = \prod_{j \geq 1} \text{sinc}(a_j t), \quad (5)$$

$$\kappa_{2k}(X_a) = \frac{2^{2k} B_{2k}}{2k} p_k(a) \quad (k \geq 1), \quad (6)$$

$$\kappa_{2k+1}(X_a) = 0 \quad (k \geq 0). \quad (7)$$

Here $\text{sinc } z = \sin z/z$, with its removable value at zero, and B_m are the Bernoulli numbers with $B_2 = 1/6$ and $B_4 = -1/30$. For $|t| < \pi/a_1$ when $a_1 > 0$,

$$\log \phi_a(t) = - \sum_{k \geq 1} \frac{\zeta(2k)}{k\pi^{2k}} p_k(a) t^{2k}. \quad (8)$$

Proof. The map from the product cube $[-1, 1]^{\mathbb{N}}$ to $\mathbb{R}$ given by $(u_j) \mapsto \sum_j a_j u_j$ is continuous: its tails are bounded uniformly by $\sum_{j > N} a_j$. Its image is the entire interval in (4), since the constant sequence $u_j = x/S(a)$ realizes any point of that interval when $S(a) > 0$. Product uniform measure gives positive measure to every nonempty basic open cylinder. Continuity therefore implies that every point of the image is in the support of its pushforward. The case $S(a) = 0$ is immediate.

Independence gives (5) first for partial sums and then by bounded convergence. Euler's product for sine yields

$$\log \text{sinc } z = - \sum_{k \geq 1} \frac{\zeta(2k)}{k\pi^{2k}} z^{2k}, \quad |z| < \pi.$$

For $|t| < \pi/a_1$, the double series is absolutely convergent. Indeed, $p_k(a) \leq a_1^{2k-2} p_1(a)$, so after extracting $t^2 p_1(a)$ the remaining series is geometrically dominated. This justifies summation over j and gives (8).

Alternatively expand $\log(\sinh z/z) = \sum_{k \geq 1} 2^{2k} B_{2k} z^{2k}/(2k(2k)!)$ near zero and use additivity of log moment-generating functions. The coefficient of $z^{2k}/(2k)!$ is (6). Symmetry gives (7). In particular, $\text{Var}(X_a) = p_1(a)/3$. $\square$

Moment and cumulant lists determine each other triangularly. Thus equality of $p_1, \dots, p_M$ is equivalent to equality of even cumulants through order $2M$, and it implies equality of all moments through degree $2M + 1$. The extra odd degree uses symmetry. Since every B_{2k} is nonzero, no even power sum disappears from the full cumulant sequence.

2.2 The known uniqueness theorem

Theorem 2.2 (Identifiability; a summable special case of Billey–Swanson). *If $a, b \in \mathcal{A}_L$ and $\mu_a = \mu_b$, then $a = b$.*

Proof. Equal laws have equal cumulants and hence equal p_k . For any nonzero spectrum let $A = a_1 > 0$. Then

$$A^{2k} \leq p_k(a) \leq A^{2k-2} p_1(a),$$

so $p_k(a)^{1/(2k)} \rightarrow A$. Its multiplicity is finite and equals

$$\lim_{k \rightarrow \infty} \frac{p_k(a)}{A^{2k}} = \sum_j \lim_{k \rightarrow \infty} (a_j/A)^{2k} = \#\{j : a_j = A\}.$$

Dominated convergence is justified by the summable majorant $(a_j/A)^2$. Remove these largest equal entries from both spectra and subtract their contribution from every power sum. Repeat. Every positive coefficient at a fixed rank is reached after finitely many such removals. If a remaining power-sum sequence vanishes, all its coefficients vanish. This recovers the whole padded decreasing sequence. $\square$

The proof is included to specify exactly what the inverse means. It does not turn the already known theorem into a new result. Nor is its limit procedure a robust numerical algorithm: taking high cumulants and successive deflations can amplify observational errors severely.

2.3 Continuity without a useful power-law modulus

Proposition 2.3. *The space $\mathcal{A}_L$ is compact in ℓ^2 . Its map to laws is continuous in the quadratic Wasserstein distance W_2 , with*

$$W_2(\mu_a, \mu_b) \leq \frac{\|a - b\|_2}{\sqrt{3}}. \quad (9)$$

It is a homeomorphism onto its image in W_2 and, equivalently on this bounded-support image, in the weak topology.

Proof. Monotonicity implies $a_j \leq L/j$. Consequently $\sum_{j>N} a_j^2 \leq L^2 \sum_{j>N} j^{-2} \leq L^2/N$, uniformly in $a \in \mathcal{A}_L$. A coordinatewise diagonal subsequence therefore converges in ℓ^2 . Its limit is decreasing and nonnegative, and has sum at most L by Fatou's lemma. This proves compactness.

Couple both random series with the same U_j . Independence and centering give $\mathbb{E}|X_a - X_b|^2 = \|a - b\|_2^2/3$, which proves (9). By [theorem 2.2](#), the continuous map is injective. A continuous injection from a compact space into a Hausdorff space is a homeomorphism onto its image. Finally, all supports are in $[-L, L]$, where weak convergence implies convergence of second moments and hence W_2 convergence. $\square$

In particular an inverse modulus tending to zero exists. The main theorem concerns its *rate*. There is no contradiction between a compact homeomorphism and a failure of every Hölder estimate. We make no claim that the forward map is globally continuous in total variation: approaching a point mass is an elementary obstruction to such a statement.

Proposition 2.4 (A largest-factor certificate). *If $a \in \mathcal{A}_L$, $L > 0$, and $k \geq 1$, then*

$$\left(\frac{p_k(a)}{L} \right)^{1/(2k-1)} \leq a_1 \leq p_k(a)^{1/(2k)}. \quad (10)$$

The length of this interval is at most $L/(2k)$.

Proof. The upper estimate uses $a_1^{2k} \leq p_k$; the lower uses $p_k \leq a_1^{2k-1} \sum_j a_j \leq L a_1^{2k-1}$. Put $q = p_k/L^{2k} \in [0, 1]$. The interval length divided by L is $q^{1/(2k)} - q^{1/(2k-1)}$. Its maximum on $[0, 1]$ is

$$\frac{1}{2k} \left(\frac{2k-1}{2k} \right)^{2k-1} \leq \frac{1}{2k},$$

by differentiation, including the endpoint values zero. $\square$

If a certified measurement satisfies $|\hat{p}_k - p_k| \leq \delta$, the same proof gives the enclosing interval

$$\left[\left(\frac{\max\{0, \hat{p}_k - \delta\}}{L} \right)^{1/(2k-1)}, (\hat{p}_k + \delta)^{1/(2k)} \right] \cap [0, L].$$

This is a statement about a *certified power-sum error*. It does not supply such an error from raw empirical moments, and it does not establish a rate for recovery of the entire spectrum.

3 The smooth dyadic reference law

Proposition 3.1. *The random variable $X_0 = \sum_{j \geq 1} 2^{-j} U_j$ has an even, unimodal density $f \in C_c^\infty(\mathbb{R})$ with*

$$\text{supp } f = [-1, 1], \quad f(0) = 1, \quad \|f'\|_1 = 2, \quad \text{Var}(X_0) = \frac{1}{9}.$$

This is the usual Rvachev up-density in the normalization of [2].

Proof. Support and variance follow from [theorem 2.1](#) and $\sum_{j \geq 1} 4^{-j} = 1/3$. For any fixed m , retaining the first m factors in the characteristic product gives

$$|\phi_0(t)| \leq \prod_{j=1}^m \min\{1, 2^j/|t|\} \leq C_m(1 + |t|)^{-m}.$$

Thus $t^r \phi_0(t)$ is integrable for every r . Fourier inversion and differentiation under the integral give a C^∞ density, necessarily compactly supported.

For completeness, even unimodality is preserved by convolution with a centered uniform density. If g is even and nonincreasing on $[0, \infty)$, then the derivative of $(2a)^{-1} \int_{x-a}^{x+a} g(u) du$ is $(g(x+a) - g(x-a))/(2a) \leq 0$ for $x \geq 0$, wherever the derivative exists. Induction applies to finite convolutions. Their characteristic functions are dominated, from the second convolution onward, by an integrable multiple of $(1 + |t|)^{-2}$. Fourier inversion therefore gives uniform convergence of their densities to f . Even unimodality passes to the limit.

Write $X_0 = U_1/2 + Y$, where Y is supported in $[-1/2, 1/2]$ and has no atoms, because it itself has an independent nondegenerate uniform summand. The first summand has density 1 on $[-1/2, 1/2]$. Its convolution with Y gives $f(0) = 1$. Smoothness, evenness, monotonicity, and compact support then yield $\|f'\|_1 = 2f(0) = 2$. $\square$

Lemma 3.2 (Translation and smoothing estimates). *If Y is independent of X_0 and integrable, then*

$$\text{TV}(\text{Law}(X_0 + Y), \mu_0) \leq \mathbb{E}|Y|. \quad (11)$$

If Y is centered and square-integrable, the right side is at most $\sqrt{\text{Var } Y}$. For an arbitrary probability measure ν ,

$$\|(f * \nu)^{(r)}\|_\infty \leq \|f^{(r)}\|_\infty \quad (r \geq 0). \quad (12)$$

Proof. The fundamental theorem of calculus and Fubini give $\|f(\cdot - y) - f\|_1 \leq |y| \|f'\|_1 = 2|y|$. Average this inequality over Y and divide by two. Cauchy–Schwarz gives the variance bound. Differentiation under convolution with a probability measure proves (12). $\square$

The role of the common Fabius factor is now clear. It makes every constructed law smooth without demanding that the perturbing finite sum be smooth to all orders, and it provides explicit total-variation control of convergence back to the reference law.

4 Exact finite-moment fibres with fixed support

4.1 A rank argument that preserves positivity

Lemma 4.1 (Generalized Vandermonde rank). *For distinct positive $t_1, \dots, t_d$ and distinct nonnegative integers $e_1 < \dots < e_d$, the matrix $(t_j^{e_i})_{i,j=1}^d$ is invertible.*

Proof. A nonzero linear combination of d distinct monomials has at most $d - 1$ distinct positive zeros. Prove this by induction on d : divide by the smallest power of t without changing positive zeros, differentiate, and apply Rolle's theorem. The derivative has at most $d - 1$ nonzero monomials, so has at most $d - 2$ positive zeros. If a linear combination of the rows vanished at all d nodes, this zero bound would be contradicted. $\square$

Theorem 4.2 (A curve of laws invisible to any prescribed finite moment list). *For every integer $M \geq 1$, there are $\delta > 0$ and a nonconstant real-analytic curve $\theta \mapsto a(\theta) \in \mathcal{A}_1$, $|\theta| < \delta$, such that $a(0) = a^0$ and:*

- (i) *only the first $M + 2$ coefficients change;*
- (ii) *$\text{supp } \mu_{a(\theta)} = [-1, 1]$ exactly;*
- (iii) *all moments through degree $2M + 1$ are independent of θ ;*
- (iv) *the cumulant of order $2M + 2$ has a nonzero derivative at zero after a suitable local parametrization;*
- (v) *the laws have C_c^∞ densities and tend to μ_0 in total variation as $\theta \rightarrow 0$.*

Proof. Set $r = M + 2$ and let $t = (t_1, \dots, t_r)$ denote the changing half-lengths. Consider the analytic map

$$F(t) = \left(\sum_{j=1}^r t_j, \sum_{j=1}^r t_j^2, \sum_{j=1}^r t_j^4, \dots, \sum_{j=1}^r t_j^{2M} \right).$$

At $t^0 = (2^{-1}, \dots, 2^{-r})$, its Jacobian has full row rank $M + 1$. After nonzero row factors are removed, any square maximal minor is a matrix from [theorem 4.1](#), with exponents $0, 1, 3, \dots, 2M - 1$.

The analytic implicit function theorem gives a one-dimensional analytic level manifold through t^0 . Positivity, strict decreasing order, and the gap above the unchanged coefficient 2^{-r-1} are open conditions, so a sufficiently short curve in this manifold remains a valid spectrum. Constancy of its first coordinate preserves $\sum_j a_j = 1$, proving (ii). Constancy of the other coordinates preserves $p_1, \dots, p_M$ and hence (iii).

Append $G(t) = \sum_j t_j^{2M+2}$ to F . The resulting square Jacobian is invertible by the same lemma, now with exponents $0, 1, 3, \dots, 2M + 1$. Thus dG does not vanish on the one-dimensional kernel of dF . We may use $G - G(t^0)$ itself as a local parameter on the level manifold. Since $B_{2M+2} \neq 0$, (6) proves (iv). In particular, the laws are genuinely different; this is not a reordering of identical factors.

The unchanged tail is distributed as $2^{-r} X_0$. It has a smooth compact support density $g(x) = 2^r f(2^r x)$. All laws on the curve are convolutions with this common density. Couple the finite changing prefixes with the same $U_1, \dots, U_r$. Their difference tends to zero in L^1 , and $\|g(\cdot - y) - g\|_1 \leq |y| \|g'\|_1$ proves total-variation convergence. Smoothness follows by convolution, completing (v). $\square$

More generally, changing r distinct positive coefficients while fixing the support radius and the first M even power sums gives a local manifold of dimension $r - M - 1$ whenever $r \geq M + 1$. In particular, $M + 2$ is the smallest number of distinct changing coefficients for which this regular local argument gives a nonconstant curve. This is a local rank statement, not a global classification of all moment fibres.

4.2 A completely explicit fixed-variance example

Example 4.3. For $M = 1$, leave $a_j = 2^{-j}$ for $j \geq 4$ and set

$$\begin{aligned} a_1(\theta) &= (7 + 5 \cos \theta - \sqrt{3} \sin \theta)/24, \\ a_2(\theta) &= (7 - \cos \theta + 3\sqrt{3} \sin \theta)/24, \\ a_3(\theta) &= (7 - 4 \cos \theta - 2\sqrt{3} \sin \theta)/24. \end{aligned}$$

For sufficiently small $|\theta|$ these entries are positive, strictly decreasing, and greater than $1/16$. Direct calculation gives

$$\sum_{j=1}^3 a_j(\theta) = \frac{7}{8}, \quad \sum_{j=1}^3 a_j(\theta)^2 = \frac{21}{64}.$$

The full support is exactly $[-1, 1]$ and the variance remains $1/9$. Nevertheless

$$\left. \frac{d}{d\theta} \kappa_4(X_{a(\theta)}) \right|_{\theta=0} = \frac{7\sqrt{3}}{3840} \neq 0.$$

Thus even exact support and exact variance do not identify the Fabius spectrum. The verification script checks these identities symbolically.

The finite-fibre theorem and the quantitative theorem below answer *different* questions. The former fixes the exact support endpoints and agrees with μ_0 on a chosen finite moment list. The latter compares two laws that are exponentially close to one another in total variation; it imposes a common support bound, not exact support endpoints ± 1 . Neither statement should be silently substituted for the other.

5 Chebyshev pairs of genuine uniform-factor spectra

5.1 Two root sets with exactly matching initial power sums

Let T_n be the Chebyshev polynomial characterized by $T_n(\cos \theta) = \cos(n\theta)$. For $n \geq 2$, list in decreasing order

$$y_0^+ > \cdots > y_{n-1}^+, \quad y_0^- > \cdots > y_{n-1}^-$$

the roots of $T_n(y) = 1/2$ and $T_n(y) = -1/2$, respectively. Every root is simple and in $(-1, 1)$. On the j th interval of monotonicity in angular coordinates, the two angles are

$$\theta_j^\pm = \frac{j\pi + \alpha_j^\pm}{n}, \quad \alpha_j^+ = \begin{cases} \pi/3 & j \text{ even}, \\ 2\pi/3 & j \text{ odd}, \end{cases} \quad \alpha_j^- = \pi - \alpha_j^+. \quad (13)$$

Then $y_j^\pm = \cos \theta_j^\pm$.

Lemma 5.1. For $1 \leq k < n$,

$$\sum_{j=0}^{n-1} (3 + y_j^+)^k = \sum_{j=0}^{n-1} (3 + y_j^-)^k. \quad (14)$$

At the first unmatched order,

$$\sum_{j=0}^{n-1} (3 + y_j^+)^n - \sum_{j=0}^{n-1} (3 + y_j^-)^n = n2^{1-n}. \quad (15)$$

Both first power sums in (14) are $3n$.

Proof. The monic polynomials

$$P_{\pm}(y) = 2^{1-n}(T_n(y) \mp 1/2)$$

differ only in their constant coefficient, with $[y^0](P_+ - P_-) = -2^{1-n}$. Newton's identities say that their root power sums agree in degrees $1, \dots, n-1$; at degree n the difference is $-n$ times this constant-coefficient difference, namely $n2^{1-n}$. The binomial theorem transports both assertions from y to $3 + y$: all lower-degree differences cancel. The coefficient of y^{n-1} in T_n vanishes, so the root sum is zero, giving $3n$ after translation. $\square$

The translated roots lie in $(2, 4)$. Taking their positive square roots therefore produces actual uniform half-lengths, not a signed measure or formal perturbation. This positivity is essential to the inverse problem.

5.2 Insertion into an empty dyadic scale interval

Define

$$k_n = \lceil \log_2 n \rceil, \quad \eta_n = 2^{-k_n-1}, \quad \frac{1}{4n} \leq \eta_n \leq \frac{1}{2n}. \quad (16)$$

Let all uniforms in the following expression be independent, and independent of X_0 :

$$S_n^{\pm} = \sum_{j=0}^{n-1} \sqrt{3 + y_j^{\pm}} U_j + \sum_{\ell=1}^{2n} V_{\ell}, \quad X_n^{\pm} = X_0 + \eta_n S_n^{\pm}. \quad (17)$$

Let $a_n^{\pm}$ be the decreasing rearrangement of the multiset

$$\{2^{-j} : j \geq 1\} \cup \{\eta_n \sqrt{3 + y_j^{\pm}} : 0 \leq j < n\} \cup \{\eta_n \text{ repeated } 2n \text{ times}\}. \quad (18)$$

The last $2n$ identical factors are not cosmetic. They suppress the large-frequency part of the Fourier transform, where moment matching alone would not control total variation.

Proposition 5.2. *The spectra in (18) belong to $\mathcal{A}_3$ and satisfy*

$$\text{Var}(X_n^{\pm}) = \frac{1}{9} + \frac{5n\eta_n^2}{3}, \quad (19)$$

$$\text{TV}(\text{Law}(X_n^{\pm}), \mu_0) \leq \sqrt{5n/3} \eta_n \leq \sqrt{5/(12n)}. \quad (20)$$

Their densities $f_n^{\pm}$ are smooth, supported in $[-3, 3]$, and satisfy $\|(f_n^{\pm})^{(r)}\|_{\infty} \leq \|f^{(r)}\|_{\infty}$ for all r . The laws agree in moments through degree $2n-1$, whereas

$$\kappa_{2n}(X_n^+) - \kappa_{2n}(X_n^-) = 2^n B_{2n} \eta_n^{2n} \neq 0. \quad (21)$$

Proof. Each of the n variable half-lengths before scaling is less than 2, so the added sum of half-lengths is at most $4n\eta_n \leq 2$. The base contributes 1. Their total squared half-length before scaling is $3n + 2n = 5n$ by theorem 5.1, proving (19). Apply theorem 3.2 with $Y = \eta_n S_n^{\pm}$ to obtain (20) and all smoothness bounds.

The first $n-1$ even power sums agree by (14); the base and the repeated factors are common. Hence all moments through degree $2n-1$ agree. Multiplying (15) by $\eta_n^{2n} 2^{2n} B_{2n}/(2n)$ gives (21). The first different cumulant is also the first different moment, since the moment-cumulant relation is triangular. $\square$

5.3 A middle coordinate gives a quadratic, not cubic, separation

Lemma 5.3 (Ordered spectral separation). *For all $n \geq 2$,*

$$d_\infty(a_n^+, a_n^-) \geq \frac{1}{48n^2}. \quad (22)$$

More precisely, put $j_n = \lfloor (n-1)/2 \rfloor$. A witnessing coordinate is $k_n + 1 + j_n$, whose absolute difference is

$$\Delta_n = \eta_n \left| \sqrt{3 + y_{j_n}^+} - \sqrt{3 + y_{j_n}^-} \right|. \quad (23)$$

Proof. All inserted variable half-lengths lie strictly between η_n and $2\eta_n$. There are exactly k_n base dyadic factors at least $2\eta_n$, and no base factors strictly between these endpoints. All repeated factors equal the lower endpoint. Therefore the ordered positions $k_n + 1, \dots, k_n + n$ consist exactly of the inserted variable factors, in their displayed order. Sorting does not erase the difference.

The two angles at position j_n have distance $\pi/(3n)$ and midpoint

$$m_n = \frac{(j_n + 1/2)\pi}{n} \in [\pi/4, \pi/2].$$

Consequently

$$|y_{j_n}^+ - y_{j_n}^-| = 2 \sin m_n \sin \frac{\pi}{6n} \geq 2 \cdot \frac{1}{2} \cdot \frac{1}{3n} = \frac{1}{3n},$$

where $\sin x \geq 2x/\pi$ for $0 \leq x \leq \pi/2$. Rationalizing the square roots in (23), their denominator is at most 4. Thus $\Delta_n \geq \eta_n/(12n) \geq 1/(48n^2)$. $\square$

Using only the largest inserted factor would yield a weaker n^{-3} bound, because cosine is flat at its endpoint. The middle coordinate avoids that loss. Along $n = 2^m$, elementary expansion of (23) even gives

$$\Delta_n \sim \frac{\pi}{12\sqrt{3}} n^{-2}.$$

This is an asymptotic for the displayed witnessing coordinate, not an assertion that the full d_∞ distance equals that coordinate.

6 From moment matching to exponential total-variation closeness

The next theorem is the analytic core. It does not use an asymptotic approximation to the density. In particular, its very small bounds are not obtained by subtracting nearly equal floating-point density values.

Theorem 6.1 (A uniform Fourier certificate). *For $n \geq 2$ and $S_n^\pm$ as in (17),*

$$\text{TV}(\text{Law}(S_n^+), \text{Law}(S_n^-)) \leq 40n(6/7)^{2n-1}. \quad (24)$$

The same bound holds for $X_n^\pm$.

Proof. Put $\rho = 6/7$, $q = 4/\pi^2 < 1/2$, and let

$$F_\pm(t) = \left(\prod_{j=0}^{n-1} \text{sinc}(\sqrt{3 + y_j^\pm} t) \right) \text{sinc}(t)^{2n}, \quad D(t) = F_+(t) - F_-(t).$$

We estimate the Fourier H^1 norm, treating small and large frequencies separately.

Small frequencies: $|t| \leq 1$. Every sinc argument has magnitude at most $2 < \pi$, so all factors are positive. Let $L_{\pm}(t) = \log F_{\pm}(t) \leq 0$ and $H = L_+ - L_-$. The common factors cancel from H . For any $k \geq 1$ the difference between the variable power sums has absolute value at most $2n4^k$; for $k < n$ it is zero. Hence

$$|H(t)| \leq 4n \sum_{k \geq n} \frac{q^k |t|^{2k}}{k} \leq \frac{4}{1-q} q^n |t|^{2n} \leq 8q^n |t|^{2n},$$

$$|H'(t)| \leq 8n \sum_{k \geq n} q^k |t|^{2k-1} \leq 16nq^n |t|^{2n-1}.$$

We used $\zeta(2k) < 2$ and the convergent log-sinc expansion; termwise differentiation is justified by geometric domination on this interval.

For $|s| \leq 2$, $\text{sinc}(s) \geq 1/3$. Indeed sinc decreases on $[0, 2]$, and the alternating sine series gives $\sin 2 \geq 2 - 2^3/6 = 2/3$. Also $|\text{sinc}'(s)| \leq \mathbb{E}|U| = 1/2$, by differentiating the characteristic function of $U \sim \text{Unif}[-1, 1]$. A variable factor therefore contributes at most 3 to $|L'_{\pm}|$, and a common unit factor at most $3/2$. Thus $|L'_{\pm}| \leq 6n$.

The identity

$$D = H \int_0^1 e^{(1-u)L_- + uL_+} du$$

is useful because the exponential is at most 1, avoiding any unstable division by $F_{\pm}$. Its derivative gives

$$|D(t)| \leq 8q^n |t|^{2n}, \quad |D'(t)| \leq 64nq^n |t|^{2n-1}. \quad (25)$$

Integrating the squares, even the coarse estimate by interval length gives

$$\int_{|t| \leq 1} (|D(t)|^2 + |D'(t)|^2) dt \leq 8320n^2 q^{2n}. \quad (26)$$

Large frequencies: $|t| \geq 1$. Let $h(t) = |\text{sinc } t| \leq 1$. All the other sinc factors have modulus at most 1. The product rule, used without dividing by any factor, gives

$$|F_{\pm}(t)| \leq h(t)^{2n}, \quad |F'_{\pm}(t)| \leq nh(t)^{2n} + nh(t)^{2n-1} \leq 2nh(t)^{2n-1}.$$

Therefore

$$|D(t)|^2 + |D'(t)|^2 \leq 20n^2 h(t)^{4n-2}. \quad (27)$$

On $1 \leq |t| \leq 2$, sinc is positive in magnitude and bounded by $\sin 1 < 6/7$. For an exact elementary bound, use $\sin 1 \leq 1 - 1/6 + 1/120 = 101/120 < 6/7$. On $|t| \geq 2$, use $h(t) \leq 1/|t|$. It follows that

$$\int_{|t| \geq 1} h(t)^{4n-2} dt \leq 2\rho^{4n-2} + \frac{2^{-(4n-4)}}{4n-3} \leq 4\rho^{4n-2}.$$

The last inequality holds for $n \geq 2$, for example by comparing $(1/2)^{4n-2}$ to ρ^{4n-2} . Combining this with (27) gives

$$\int_{|t| \geq 1} (|D(t)|^2 + |D'(t)|^2) dt \leq 80n^2 \rho^{4n-2}. \quad (28)$$

Since $q^{2n} \leq \rho^{4n-2}$,

$$\int_{\mathbb{R}} (|D|^2 + |D'|^2) dt \leq 8400n^2 \rho^{4n-2}. \quad (29)$$

Conversion to total variation. Let g be the difference of the densities of S_n^+ and S_n^- . These finite sums have bounded compactly supported densities, so $g, xg \in L^2$. For the Fourier convention $\widehat{g}(t) = \int e^{itx} g(x) dx = D(t)$, Cauchy–Schwarz and Plancherel yield

$$\begin{aligned} \|g\|_1 &\leq \left(\int_{\mathbb{R}} \frac{dx}{1+x^2} \right)^{1/2} \left(\int_{\mathbb{R}} (1+x^2) |g(x)|^2 dx \right)^{1/2} \\ &= \frac{1}{\sqrt{2}} \left(\int_{\mathbb{R}} (|D(t)|^2 + |D'(t)|^2) dt \right)^{1/2}. \end{aligned}$$

Dividing by two and using (29) proves

$$\text{TV}(\text{Law}(S_n^+), \text{Law}(S_n^-)) \leq \frac{\sqrt{8400}}{2\sqrt{2}} n\rho^{2n-1} < 40n\rho^{2n-1}.$$

Total variation is invariant under a common nonzero dilation and contracts under convolution with a common probability measure. Apply these facts first with dilation η_n and then convolution by μ_0 to obtain the assertion for $X_n^\pm$. $\square$

Remark 6.2 (Why both frequency ranges are necessary). Equal low-order moments alone control only a neighborhood of frequency zero. A proof that stopped with (25) would not establish closeness of densities in L^1 . The common $2n$ unit factors create the large-frequency estimate (28). Keeping a derivative of the Fourier difference is also essential to the particular weighted Cauchy–Schwarz argument used here. These two features explain the architecture of the construction.

7 Fixing the variance without sacrificing smoothness

Define the common scale

$$c_n = (1 + 15n\eta_n^2)^{-1/2}, \quad \tilde{X}_n^\pm = c_n X_n^\pm, \quad \tilde{a}_n^\pm = c_n a_n^\pm. \quad (30)$$

Proposition 7.1. *The spectra $\tilde{a}_n^\pm$ belong to $\mathcal{K}$ and satisfy all the quantitative assertions of theorem 1.2 preceding its non-Hölder conclusion. Their first unmatched cumulant is*

$$\kappa_{2n}(\tilde{X}_n^+) - \kappa_{2n}(\tilde{X}_n^-) = c_n^{2n} 2^n B_{2n} \eta_n^{2n} \neq 0.$$

Proof. Equation (19) shows that the common variance becomes $1/9$. Since $15n\eta_n^2 \leq 15/(4n) \leq 15/8$, we have $1/2 < c_n \leq 1$. Supports remain in $[-3, 3]$, and the scaled densities satisfy

$$\left\| (\tilde{f}_n^\pm)^{(r)} \right\|_\infty = c_n^{-r-1} \left\| (f_n^\pm)^{(r)} \right\|_\infty \leq 2^{r+1} \left\| f^{(r)} \right\|_\infty.$$

The spectral distance is multiplied by c_n , proving (1). Total variation between the pair is unchanged, proving (2). Equal moments remain equal under common scaling, and cumulants scale homogeneously, giving the displayed defect.

It remains to control the distance to the original reference law rather than to its dilation. Write $f_c(x) = c^{-1} f(x/c)$. Direct differentiation and a change of variables give

$$\|\partial_c f_c\|_1 = c^{-1} \int_{\mathbb{R}} |f(u) + uf'(u)| du \leq \frac{3}{c},$$

because $\|f\|_1 = 1$, $|u| \leq 1$ on the support, and $\|f'\|_1 = 2$. Therefore

$$\text{TV}(\text{Law}(cX_0), \mu_0) \leq \frac{3}{2} \log(1/c), \quad 0 < c \leq 1.$$

Using common-dilation invariance in the first term of the triangle inequality,

$$\begin{aligned} \text{TV}(\text{Law}(c_n X_n^\pm), \mu_0) &\leq \text{TV}(\text{Law}(X_n^\pm), \mu_0) + \frac{3}{2} \log(1/c_n) \\ &\leq \sqrt{\frac{5}{12n}} + \frac{3}{4} \log(1 + 15n\eta_n^2) \\ &\leq \sqrt{\frac{5}{12n}} + \frac{45}{16n}. \end{aligned}$$

This proves (3). $\square$

Thus instability is not explained by increasing variance, expanding support, or loss of smoothness. All three potential explanations have been removed. The theorem does allow an increasing number of small factors in one dyadic scale interval; it is this growing unresolved block that the inverse must disentangle.

8 Inverse moduli and statistical lower bounds

8.1 The exact local statement

For $r > 0$ define

$$\omega_r(\varepsilon) = \sup \left\{ d_\infty(a, b) : \begin{array}{l} a, b \in \mathcal{K}, \quad \text{TV}(\mu_a, \mu_0) < r, \quad \text{TV}(\mu_b, \mu_0) < r, \\ \text{TV}(\mu_a, \mu_b) \leq \varepsilon \end{array} \right\}. \quad (31)$$

Corollary 8.1. *For every $r > 0$, for all sufficiently small $\varepsilon > 0$,*

$$\omega_r(\varepsilon) \geq \frac{1}{11616[\log(100/\varepsilon)]^2}. \quad (32)$$

In particular no pairwise Hölder inverse estimate holds on the indicated neighborhood, for any positive exponent.

Proof. The non-Hölder assertion already follows by substituting the pairs: $1/(96n^2)$ eventually exceeds $C[40n(6/7)^{2n-1}]^\alpha$, whereas both laws approach μ_0 . For the explicit modulus take

$$n = \lceil 10 \log(100/\varepsilon) \rceil, \quad 0 < \varepsilon \leq 1.$$

The elementary bound $2 \log(7/6) > 0.3$ implies

$$(6/7)^{2n-1} \leq (7/6)e^{-0.3n} \leq (7/6)(\varepsilon/100)^3.$$

Also $n \leq 51/\varepsilon$, since $\log(100/\varepsilon) \leq 5/\varepsilon$ on this range. It follows that $40n(6/7)^{2n-1} \leq 0.00238\varepsilon^2 \leq \varepsilon$. For sufficiently small ε , (3) puts both laws in the required r -ball. Finally $n \leq 11 \log(100/\varepsilon)$, so (1) gives (32). $\square$

Remark 8.2 (Pairwise local versus pointwise at the reference law). The conclusion rules out a bound that must hold for *every pair* in a neighborhood. It does not rule out a pointwise estimate

$$d_\infty(a, a^0) \leq C \text{TV}(\mu_a, \mu_0)^\alpha.$$

Our construction makes $\mu_{a_n^+}$ and $\mu_{a_n^-}$ exponentially close to each other, but the proved bound on their individual distances to μ_0 is only polynomial. Those are different comparisons.

8.2 A lower bound for recovery from samples

An estimator based on N observations is a measurable function of $(Z_1, \dots, Z_N)$ with values in the space of bounded real sequences. The loss is $d_\infty(\hat{a}, a)$. We do not require the estimator's output to be a valid uniform-factor spectrum, so the lower bound also applies to constrained estimators.

Lemma 8.3 (Two-point testing bound). *If parameters a, b have distance d and observation laws P, Q , then for every estimator,*

$$\max\{\mathbb{E}_P d_\infty(\hat{a}, a), \mathbb{E}_Q d_\infty(\hat{a}, b)\} \geq \frac{d}{4}(1 - \text{TV}(P, Q)).$$

For product laws, $\text{TV}(P^{\otimes N}, Q^{\otimes N}) \leq N \text{TV}(P, Q)$.

Proof. Choose the parameter closer to the estimator's output as a binary test. On a testing error the estimation loss is at least $d/2$, by the triangle inequality. The sum of the two testing error probabilities is at least $1 - \text{TV}(P, Q)$, as follows directly by writing the test's acceptance set in the definition of total variation. The larger risk is at least half their sum, proving the first assertion. For the second, expand the difference of product measures as a telescoping sum and use that taking a product with a probability measure does not change total variation. $\square$

Theorem 8.4 (Logarithmic minimax obstruction). *Let $N \geq 1$ and put $n_N = \lceil 10 \log(100N) \rceil$. Then*

$$\inf_{\hat{a}} \sup_{a \in \mathcal{K}} \mathbb{E}_{\mu_a^{\otimes n_N}} d_\infty(\hat{a}, a) \geq \frac{1}{768n_N^2}. \quad (33)$$

For every fixed $r > 0$, the same lower bound holds for all sufficiently large N when the supremum is restricted to $\{a \in \mathcal{K} : \text{TV}(\mu_a, \mu_0) < r\}$. Consequently no uniform bound of order $N^{-\beta}$ with $\beta > 0$ is possible on any such local class.

Proof. Use the two normalized spectra of degree n_N . The estimate from the previous proof gives

$$N \cdot 40n_N(6/7)^{2n_N-1} \leq \frac{(40)(7/6)(51)}{10^6 N} < \frac{1}{2},$$

where $n_N \leq 51N$. Thus the product observation laws have total variation at most $1/2$. Substitute their spectral gap $1/(96n_N^2)$ into [theorem 8.3](#), obtaining $1/(768n_N^2)$. The local assertion follows from [\(3\)](#). Since $n_N = O(\log N)$, the stated exclusion of algebraic rates follows. $\square$

This is an information-theoretic obstruction. It applies before any restriction on running time, numerical precision, or model architecture. It is also only a lower bound: the article does not construct an estimator attaining the logarithmic rate, nor prove that its exponent is optimal.

9 Where stable recovery does become possible

The negative theorem should not be read as saying that every finite uniform-convolution problem is equally difficult. Its bad pairs use an increasing number of unknown, closely spaced factors. Fixing their number and separating their squared lengths changes the situation.

9.1 Known tails and a finite set of unknown factors

Proposition 9.1 (Finite-factor analytic inversion). *Suppose the entire tail is known and the unknown part consists of exactly r distinct positive half-lengths $a_1, \dots, a_r$. After subtracting the tail contributions, the first r even power sums determine the unordered unknown multiset exactly. The inverse is locally real-analytic at every configuration with distinct squared half-lengths.*

Proof. Write $x_j = a_j^2$ and $s_k = \sum_{j=1}^r x_j^k$. Newton's recursion

$$e_0 = 1, \quad ke_k = \sum_{i=1}^k (-1)^{i-1} e_{k-i} s_i \quad (1 \leq k \leq r)$$

constructs the coefficients of

$$P(z) = \prod_{j=1}^r (z - x_j) = z^r - e_1 z^{r-1} + e_2 z^{r-2} - \dots + (-1)^r e_r.$$

Its roots give the multiset. The Jacobian of $(x_1, \dots, x_r) \mapsto (s_1, \dots, s_r)$ has determinant

$$r! \prod_{1 \leq i < j \leq r} (x_j - x_i),$$

up to the harmless sign determined by ordering. It is nonzero at distinct nodes. The inverse function theorem, followed by positive square roots, gives the local analytic inverse. $\square$

Proposition 9.2 (An explicit root-location certificate). *Suppose the squared half-lengths satisfy $m^2 \leq x_i \leq A^2$ and $|x_i - x_j| \geq \gamma > 0$ for $i \neq j$. Let*

$$P(z) = z^r + \sum_{j=0}^{r-1} c_j z^j, \quad Q(z) = z^r + \sum_{j=0}^{r-1} \hat{c}_j z^j.$$

Choose $0 < h < \min\{\gamma/2, m^2/2\}$. If

$$\sum_{j=0}^{r-1} |\hat{c}_j - c_j| (A^2 + h)^j < h(\gamma - h)^{r-1}, \quad (34)$$

then Q has exactly one root, counting multiplicity, in each disk $|z - x_i| < h$. If its coefficients are real, all these roots are real and positive. The corresponding half-lengths are within h/m of the true ones after matching.

Proof. On $|z - x_i| = h$, one factor of P has modulus h and all others have modulus at least $\gamma - h$. The left side of (34) bounds $|Q - P|$ on that circle. Rouché's theorem gives one root of Q in each of the disjoint disks. Real coefficients force nonreal roots to occur in conjugate pairs; a disk centered on the real axis containing just one root cannot contain a nonreal pair. Its real root is positive since $h < m^2/2$. If $\hat{x}_i$ is that root, then

$$|\sqrt{\hat{x}_i} - \sqrt{x_i}| = \frac{|\hat{x}_i - x_i|}{\sqrt{\hat{x}_i} + \sqrt{x_i}} < \frac{h}{m}.$$

$\square$

These are standard finite-dimensional mechanisms, included as a contrast rather than as new root-perturbation theory. Their constants deteriorate when nodes collide or their number grows. Both problems are present in the bad Chebyshev blocks.

9.2 What a certified computation would need

A defensible finite-factor recovery procedure first specifies the number of unknown factors, a lower bound on their lengths, a separation condition, and a quantitative tail model. It then converts certified moment intervals to cumulant intervals, subtracts the tail power-sum intervals, applies Newton recursion with outward error bounds, and proves root localization by a condition such as (34). Merely finding numerical roots of a polynomial is not a certificate.

A useful elementary tail estimate is available without a detailed tail formula. If its largest half-length is at most η and its total half-length is at most L , then

$$\sum_{j \text{ in tail}} a_j^{2k} \leq L\eta^{2k-1}. \quad (35)$$

The proof factors out η^{2k-1} and sums a_j . This can be treated as an additional nonnegative uncertainty interval. It is not legitimate to discard the tail because it contributes little variance: high-order root recovery can remain sensitive to its structured contribution.

There is also a distinction between a fixed number of large factors and the whole ordered sequence. The witnessing index in [theorem 5.3](#) tends to infinity. Consequently our lower bound does not, by itself, rule out much better rates for estimating the first fixed r half-lengths under suitable tail assumptions.

10 Reproducible verification and proof boundaries

10.1 What the included program actually verifies

The package contains `code/verify.py`, requiring only Python, SymPy, and mpmath. It uses integer Chebyshev recurrences and rational Newton identities, avoiding numerical root finding for all algebraic claims. A completed run records:

Check	Count
Exact matching root power sums, before and after translation, $2 \leq n \leq 80$	6,320
Exact first unmatched power sums in both coordinates	158
Exact Bernoulli cumulant defects, $2 \leq n \leq 40$	39
Exact Jacobian rank cases and augmented determinant cases, $1 \leq M \leq 8$	8 each
Explicit analytic curve invariants and fourth-cumulant derivative	passed
90-digit Chebyshev root residual diagnostics, $2 \leq n \leq 100$	10,098
90-digit middle-coordinate gap diagnostics	99
90-digit small-frequency inequality diagnostics	360

The output files record software versions and the distinction between exact checks and floating-point diagnostics. They deliberately do not describe 90-digit arithmetic as interval certification.

The finite tests do not prove an assertion for all n . Those proofs are [theorems 5.1, 5.3](#) and [6.1](#); the program is a reproducibility and regression aid. In particular, the very small TV bounds are evaluated from a proved formula rather than estimated by numerical quadrature of a near-zero difference.

10.2 Numerical scale of the construction

For instance, the degree-256 fixed-variance pair has a displayed coordinate separation about 2.29×10^{-6} while the theorem bounds its total variation by about 6.32×10^{-31} . This does not

n	Witness Δ_n	Fixed-variance witness $c_n \Delta_n$	Proved TV upper bound
16	5.781×10^{-4}	5.204×10^{-4}	1
32	1.462×10^{-4}	1.384×10^{-4}	7.755×10^{-2}
64	3.674×10^{-5}	3.571×10^{-5}	8.054×10^{-6}
128	9.206×10^{-6}	9.074×10^{-6}	4.343×10^{-14}
256	2.304×10^{-6}	2.287×10^{-6}	6.317×10^{-31}

Table 1: Witnessing coordinate gaps, not claimed exact d_∞ distances. The gap columns are rounded evaluations of explicit expressions; the last column is the rounded value of the analytic upper bound ε_n , not an estimate of the actual total-variation distance.

mean that the actual total variation has been computed to that accuracy. It means that an exact analytic estimate certifies an upper bound of that size. The constants were chosen for a transparent proof, not numerical sharpness.

10.3 A possible Lean formalization route

The results do not inherit formal status from the repository’s existing Fabius theory. A future formalization can nevertheless be decomposed into clear interfaces. The first layer is finite algebra: Chebyshev recurrence, the level-set root description, Newton identities, translated power sums, and the middle-coordinate separation. These lemmas can be developed without probability or infinite products.

The second layer represents the finite perturbations as product-measure random variables and proves their moment and cumulant identities. The third layer isolates the Fourier estimate as a statement about two finite products, with separate small- and large-frequency lemmas and a Plancherel-to- L^1 conversion. Only then is the common Fabius factor introduced through a theorem interface giving its density, support, variance, and derivative norms. The final layer is purely metric and statistical: convolution contraction, dilation, local moduli, and the two-point bound.

This dependency order keeps a large imported Fabius development out of the finite algebraic core. It also makes the actual trust boundary visible. No Lean source or uncompiled skeleton is supplied here, and no machine-checking claim is made for the article.

11 Further research questions

Research question 1 (The sharp local inverse modulus). Determine the asymptotic order of $\omega_r(\varepsilon)$ in (31). Does a logarithmic upper bound hold under the present support, variance, and derivative constraints, and what exponent is optimal? Our result proves only $\omega_r(\varepsilon) \gtrsim (\log(1/\varepsilon))^{-2}$. Compactness gives convergence to zero without a quantitative formula. A useful next step is to combine truncated power-sum inversion with a uniform bound on the number of coefficients above a chosen threshold.

Research question 2 (Pointwise recovery of the exact dyadic spectrum). Decide whether a power-law estimate relative to the single reference parameter a^0 can hold on a natural restricted class. The dyadic spectrum has a strong separation pattern that the inserted Chebyshev blocks destroy. A proof or counterexample must compare one law directly to μ_0 , not merely compare two laws that separately approach it; [theorem 8.2](#) explains why the present argument is insufficient.

Research question 3 (Exact support endpoints together with quantitative instability). Construct explicit exponentially close pairs that retain $\text{supp } \mu = [-1, 1]$ and variance $1/9$,

approach μ_0 , and have a quantified spectral separation. [Theorem 4.2](#) preserves the exact support but does not give an exponentially small pairwise TV bound; [theorem 1.2](#) gives that bound but only a common support containment. A promising route is a small Chebyshev tail block with two compensating large-factor adjustments, while controlling the errors those adjustments introduce into low-order cumulants.

Research question 4 (Geometrically separated classes). Assume $a_{j+1} \leq qa_j$ for some fixed $q < 1$, perhaps together with bounds on the leading scale. Determine whether the inverse has a substantially better modulus. The present construction is excluded, since many factors are packed into one scale interval. Exact identifiability by successive largest-power limits suggests an approach, but converting it into certified finite-noise bounds requires controlling every deflation error.

Research question 5 (Fixed leading factors versus the complete sequence). For fixed r , find sharp rates for the loss $\max_{1 \leq j \leq r} |\hat{a}_j - a_j|$, with explicit assumptions on the remaining tail. The middle-coordinate witness here occurs at a rank growing with n , so it does not settle that finite-prefix question. It is important to distinguish a separated finite-dimensional parameter from an unknown infinite tail that can imitate its cumulants.

Research question 6 (Matching statistical upper bounds). Construct a regularized estimator with a proved risk bound over a specified infinite-factor class, and compare it to [theorem 8.4](#). Empirical high moments become unreliable quickly; a characteristic-function estimator with low-frequency regularization may be more natural. Any proposed upper rate must account for both moment cancellation and the number of unresolved small factors.

Research question 7 (Gaussian components and square-summable spectra). Extend the quantitative question to the larger generalized-uniform and Gaussian-convolution parameter spaces studied by Billey–Swanson. Their topology records the possibility that variance escapes into a Gaussian component [4]. Without an ℓ^1 support bound, new scale choices are possible, so the exponent and even the most appropriate parameter metric may change. Establishing uniqueness alone would repeat known work; a useful extension must supply quantitative stability or instability information.

Editorial note (ProveIt, 2026-09-29). For the Gaussian-variance functional, the later repository article [inverse-and-sampling/Gaussian_Dust_Christoffel_Recovery_Uniform_Factors/](#) (filed 2026-09-29, unreviewed) answers this question in a model with a known smooth compactly supported background: over square-summable spectra with total latent variance at most V , the exact minimax risk for the Gaussian variance is $V/2$ in absolute error at every sample size, and a compact class admits uniformly consistent estimation exactly when its squared-scale tails vanish uniformly. Quantitative stability of the scales themselves in this larger space is not treated there.

Research question 8 (Exact certificates for numerical inversion). Implement a validated finite-prefix recovery algorithm based on [theorems 2.4](#) and [9.2](#) and [\(35\)](#), then formalize its algebraic certificate checker. The key engineering target is an explicit statement of the measurement error model and tail assumptions, not simply a high-precision reconstruction on selected examples. This would turn the negative theorem into a practical guide for what a reliable ProveIt inverse API must require.

12 Conclusion

The complete law of a summable uniform convolution uniquely encodes its ordered factor lengths. That qualitative fact coexists with two strong obstructions. A prescribed finite list

of moments leaves nontrivial analytic families even when the support is fixed exactly. More quantitatively, a growing block of Chebyshev-derived positive factors can hide polynomial parameter differences behind exponentially small errors in the full probability law.

The construction retains fixed variance, compact support bounds, and uniform control of every density derivative. Its instability therefore lies in the factorization, not merely in rough densities or uncontrolled scaling. The explicit proofs and algebraic checks support independent review and eventual formalization. Sharp moduli, pointwise stability at the dyadic law, and matching statistical algorithms remain the central next questions.

A A compact audit of the main constants

The following chain contains all numerical losses used in the main construction:

$$\begin{aligned}
\frac{1}{4n} &\leq \eta_n \leq \frac{1}{2n}, & |y_{j_n}^+ - y_{j_n}^-| &\geq \frac{1}{3n}, \\
\Delta_n &\geq \frac{\eta_n}{12n} \geq \frac{1}{48n^2}, & \frac{1}{2} < c_n &\leq 1, \\
\int_{|t| \leq 1} (|D|^2 + |D'|^2) dt &\leq 8320n^2 q^{2n}, \\
\int_{|t| \geq 1} (|D|^2 + |D'|^2) dt &\leq 80n^2 \rho^{4n-2}, \\
q = 4/\pi^2 &< 1/2, & \rho = 6/7, & q^{2n} \leq \rho^{4n-2}, \\
\text{TV} &\leq \frac{\sqrt{8400}}{2\sqrt{2}} n \rho^{2n-1} < 40n \rho^{2n-1}.
\end{aligned}$$

The normalization changes the gap lower bound by at most a factor of two and leaves the TV bound unchanged. The statistical argument loses a further factor of eight: four from the testing reduction and two from the product-TV bound $1 - \text{TV} \geq 1/2$. Thus its denominator is $96 \cdot 8 = 768$.

The local-modulus denominator is $96 \cdot 11^2 = 11616$. Its threshold in ε depends on the prescribed neighborhood radius only through the explicit convergence estimate (3). None of these constants is claimed to be optimal.

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